SPECTRAQUANT

From Budget-Aware Allocation Failure to Verified Clipping Transfer

A reproducible study of local compression objectives and whole-model language quality

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Abstract

SpectraQuant asks whether inexpensive local compression objectives can preserve autoregressive language-model quality under a measured equal-byte checkpoint budget. The original sensitivity-aware low-rank/low-bit allocator failed against uniform INT4. A deployment-matched correction and a joint-loss diagnosis strengthened that negative result. A simpler clipping control then improved one fresh C4 holdout, motivating a frozen two-model/two-domain transfer study. The primary control improved only 2/4 cells and failed its consistency gate. Predeclared activation-weighted controls reduced C4 perplexity by 7.39-7.70% on SmolLM2-360M and 5.75-6.09% on Pythia-410M at identical checkpoint bytes, while regressing on Pythia PG19. All 52 expected Phase-4 artifacts and 42 checkpoint files were authenticated. The supported outcome is a bounded C4 improvement with a documented transfer failure, not a universal compression claim.

Headline result

Activation-weighted INT4 controls lower reserved C4 perplexity on both tested architectures at identical storage. The Pythia PG19 regression defines the boundary of that result.

Keywords: transformer compression; post-training quantization; INT4; clipping; sensitivity; low-rank approximation; reproducibility; negative results

1. Introduction

Compression methods often optimize local quantities: weight reconstruction error, activation-weighted error, Hessian surrogates, spectral statistics, or layer-wise calibration loss. The operational outcome is global. Every changed layer participates in one autoregressive computation, and locally favorable changes can interact poorly when composed.

> **Research question**
>
> Can cheap local compression scores choose a low-bit representation that preserves end-to-end language quality at the same measured checkpoint budget?

The project treats a failed original method as the start of the investigation. It first tests a joint low-rank/low-bit allocator, corrects evaluation to the actual packed operator, diagnoses marginal-versus-joint loss, simplifies the intervention to conventional clipping, and finally freezes a transfer study before holdout access.

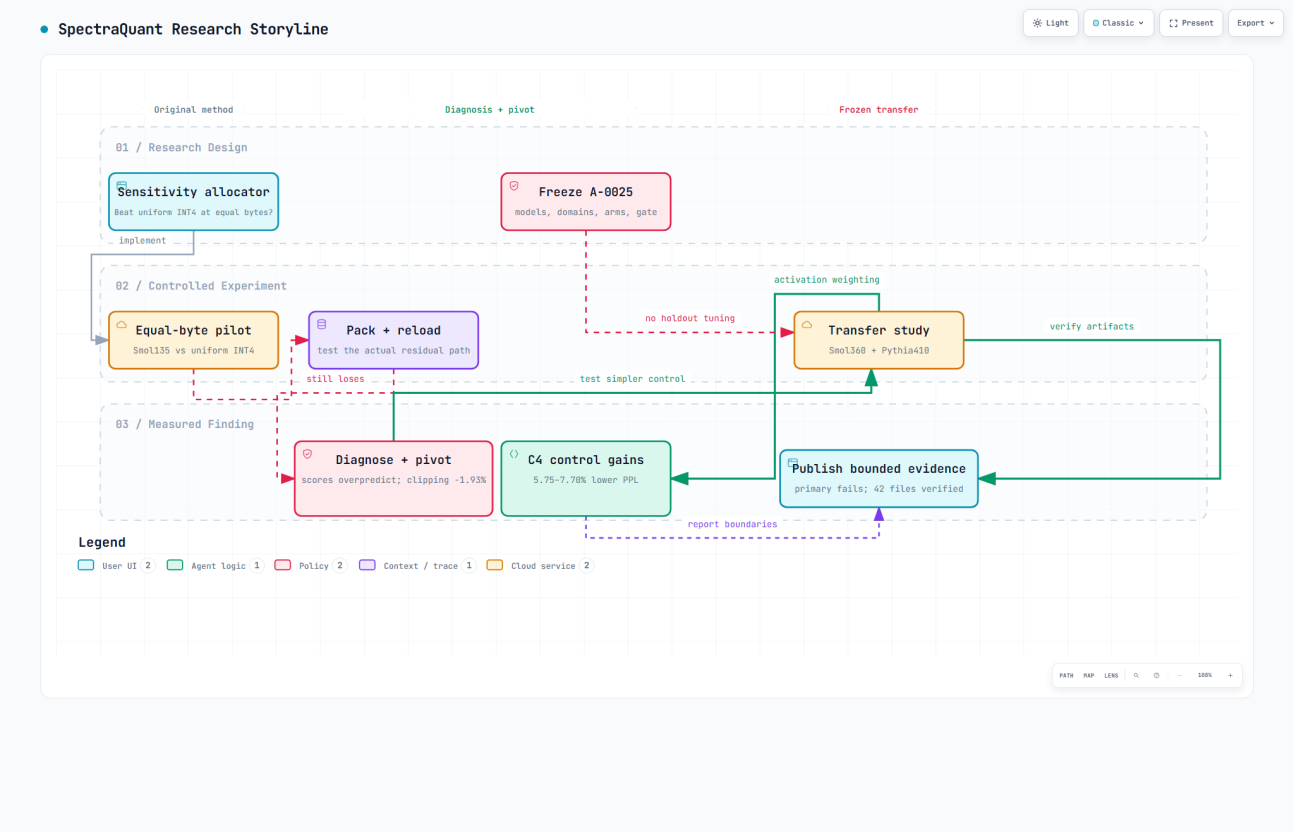


Figure 1. *The coherent progression. Each later phase tests an assumption exposed by the preceding failure.*

2. Related work and research gap

GPTQ, AWQ and SmoothQuant show that calibration statistics and second-order or activation-aware objectives can materially change low-bit behavior. HAWQ-style methods allocate precision from sensitivity estimates. LoftQ and LR-QAT combine low-rank structure with quantization-aware optimization. These results motivate the original SpectraQuant hypothesis, but they do not remove the need for equal-storage, deployment-matched comparisons.

SpectraQuant does not claim a new state-of-the-art PTQ algorithm. Its strongest contribution is an evaluation result: local proxy improvements and lower weight reconstruction error can fail to predict whole-model perplexity, while clipping benefits can depend on model, corpus and grouping orientation.

3. Problem formulation

For layer l with weight W(l), the original allocator chooses rank r(l) and bit width b(l) to minimize the sum of estimated local errors subject to a frozen byte budget:

> **Budgeted allocation objective**
>
> min sum_l E_hat(l, r(l), b(l)) subject to sum_l bytes(l, r(l), b(l)) <= B

The deployed residual representation is Q(W) + Q(B)Q(A), with rank zero representing plain quantization. Later clipping experiments keep INT4/group32 fixed and select a clipping scale from a predeclared grid. The diagonal activation proxy weights each input feature's squared weight error by its calibration second moment. It omits cross-feature covariance and is not measured output error.

4. Research and evidence architecture

Three rules govern all headline comparisons: compare quality only at equal measured bytes; reload the representation actually written to disk; and freeze models, data, arms and gates before fresh holdout access. Protocol changes are append-only. Failed gates remain visible.

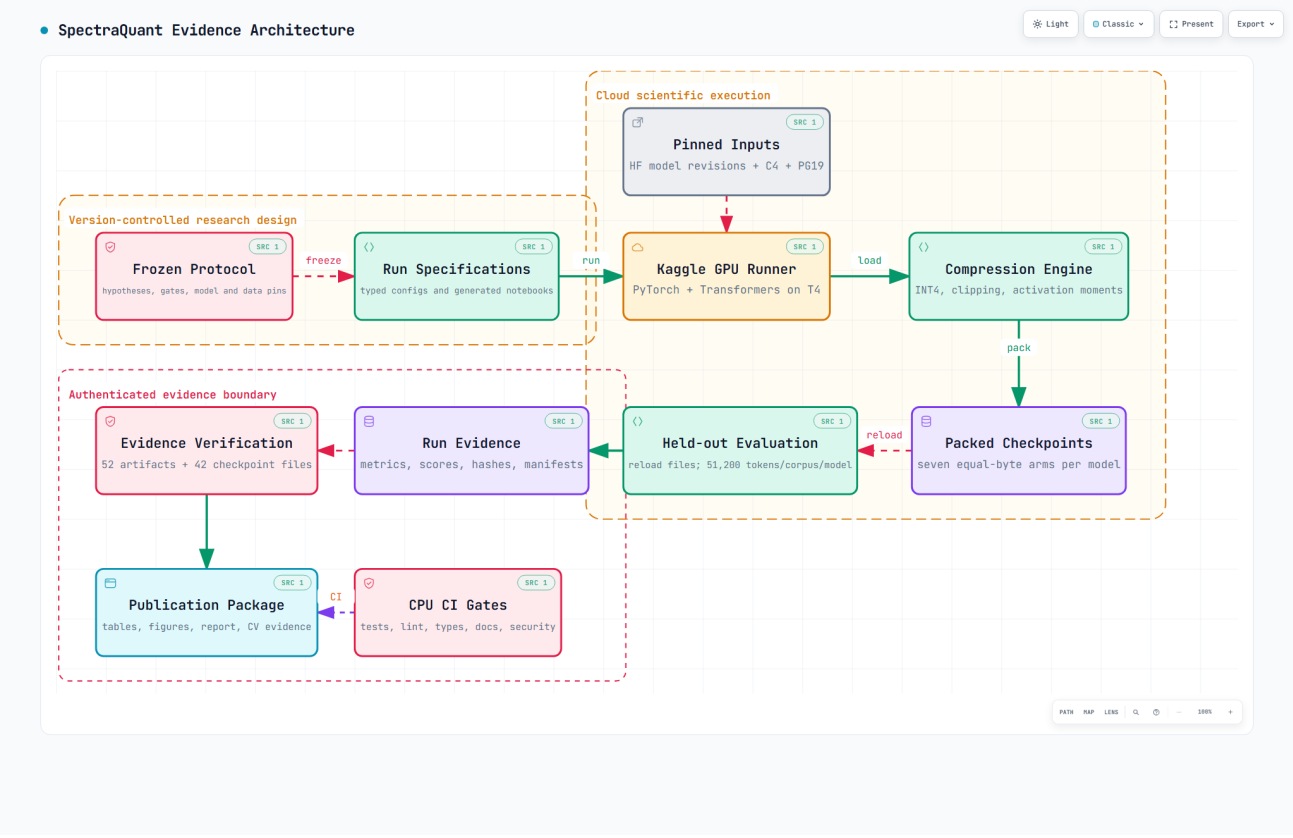


Figure 2. *Frozen design and typed run specifications feed cloud execution; packed checkpoints are reloaded for evaluation; manifests, hashes and CI gate publication.*

5. Experimental progression

Stage	Question	Measured answer
Original pilot	Beat uniform INT4 at equal bytes?	No: 23.1698/23.3131 PPL vs 18.6193.
LR-QAT direction	Reproduce the expected direction?	21.8% gap closure; below the 25% gate.
Phase 2	Does packed/reloaded evaluation rescue allocation?	No: six arms are 5.52-14.75% worse.
Phase 3 diagnosis	Do marginal gains predict joint loss?	No: all candidates overpredict and regress.
Phase 3 pivot	Can fixed clipping help once?	Yes: 28.0056 to 27.4656 PPL (-1.9279%).
Phase 4	Does the control transfer?	Mixed: primary 2/4; strong C4 secondary result.

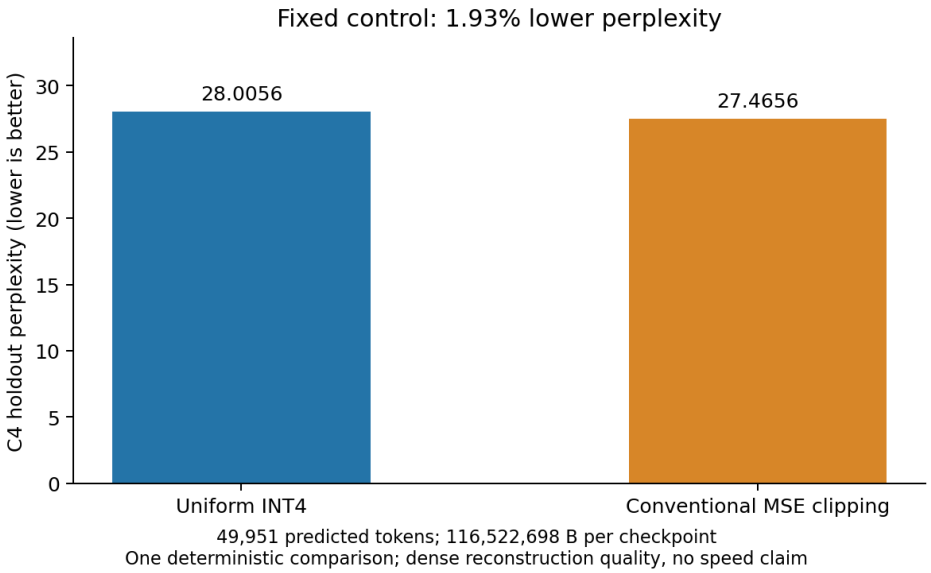


Figure 3. *The first fixed-clipping C4 holdout comparison. This bounded result triggered a frozen transfer study instead of immediate generalization.*

6. Frozen Phase-4 protocol

- **Models.** SmolLM2-360M and Pythia-410M at immutable revisions.
- **Corpora.** Reserved C4 validation documents and PG19 test-book prefixes.
- **Calibration.** The first 200 C4 training documents; no holdout NLL tuning.
- **Arms.** Seven INT4/group32 checkpoints per model: RTN/MSE on axis0/axis1 and three activation-weighted axis1 calibration subsets.
- **Evaluation.** Exactly 51,200 predicted tokens per corpus/model with document boundaries preserved.
- **Order.** All checkpoints are frozen before holdout loading.
- **Uncertainty.** Document-cluster bootstrap intervals describe corpus uncertainty; calibration subsets are not training seeds.

7. Primary transfer result

Model	Corpus	RTN PPL	MSE PPL	Reduction
SmolLM2-360M	C4	21.8140	22.3901	-2.6408%
SmolLM2-360M	PG19	16.6734	16.8819	-1.2501%
Pythia-410M	C4	32.3251	31.3777	+2.9311%
Pythia-410M	PG19	27.9244	27.0500	+3.1313%

Primary verdict: FAIL

The geometric-mean perplexity reduction is 0.5752%, above the 0.5% magnitude threshold. Only 2/4 cells improve, below the required 3/4 consistency gate.

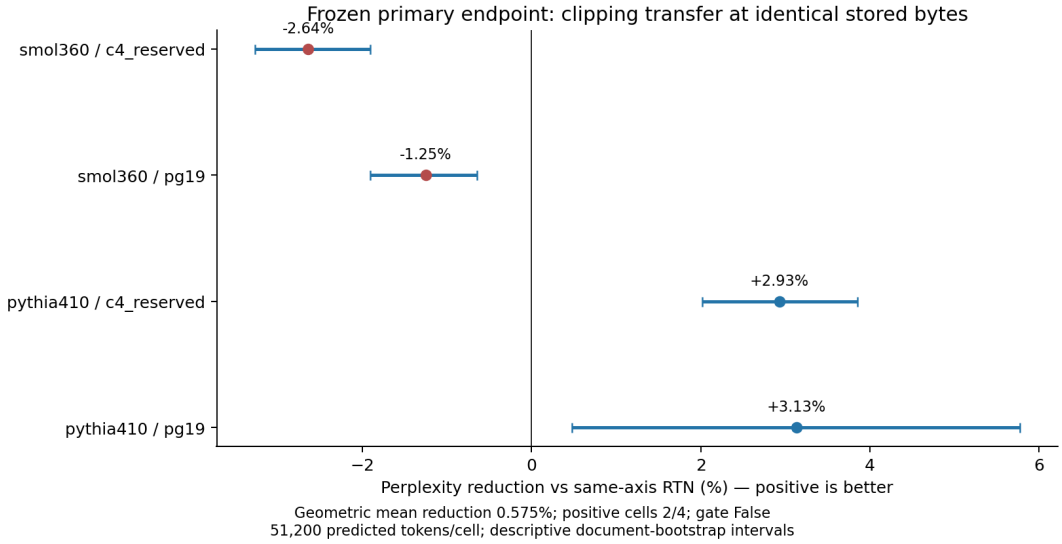


Figure 4. *Primary MSE-clipping transfer cells and descriptive intervals.*

8. Secondary activation-weighted controls

Model	Corpus	PPL reduction range	Full checkpoint
SmolLM2-360M	C4	+7.3882 to +7.7028%	271,561,031 B
SmolLM2-360M	PG19	+5.8906 to +6.1849%	271,561,031 B
Pythia-410M	C4	+5.7462 to +6.0939%	302,615,137 B
Pythia-410M	PG19	-2.5608 to -1.0864%	302,615,137 B

The control improves reserved C4 on both architectures and SmolLM2 PG19, but regresses on Pythia PG19. Calibration-domain sensitivity is plausible, but the study does not isolate it causally because it does not add book-domain calibration. The supported claim is a measured C4 improvement at equal bytes.

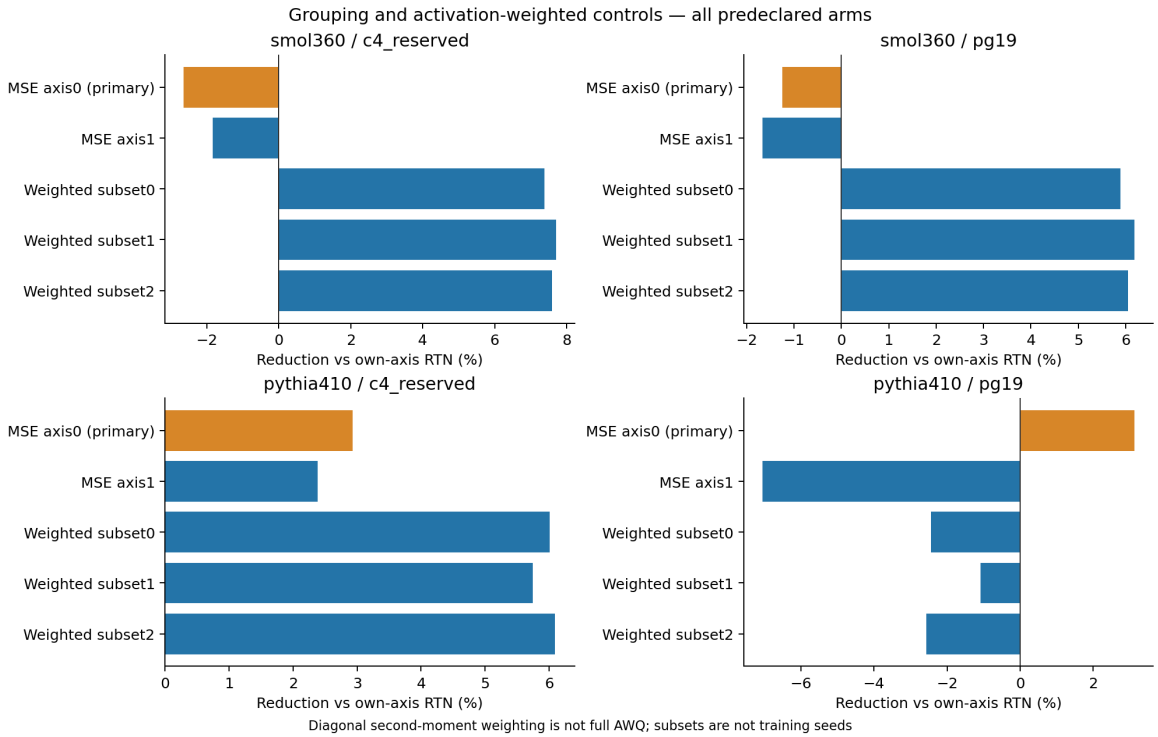


Figure 5. *All predeclared clipping controls. Secondary outcomes do not rescue the failed primary gate.*

9. Why the local objectives failed

Weight-MSE clipping reduced aggregate weight squared error by 9.98% on SmolLM2 axis0 and 10.19% on axis1, yet worsened perplexity on both corpora. Pythia axis1 provides another counterexample: 10.69% lower weight error accompanied 7.06% worse PG19 perplexity. Lower reconstruction error is therefore insufficient evidence for better language quality.

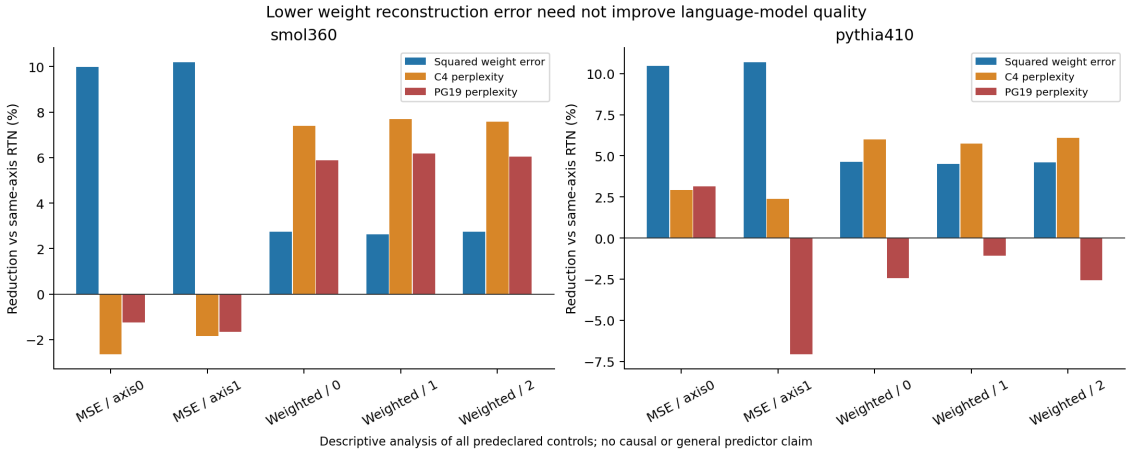


Figure 6. *Improvements in local compression objectives versus measured whole-model perplexity.*

The allocation diagnosis exposes the same problem. Layer effects measured in isolation were not additive after simultaneous deployment. Residual paths, normalization, activation shifts and error propagation couple the layers.

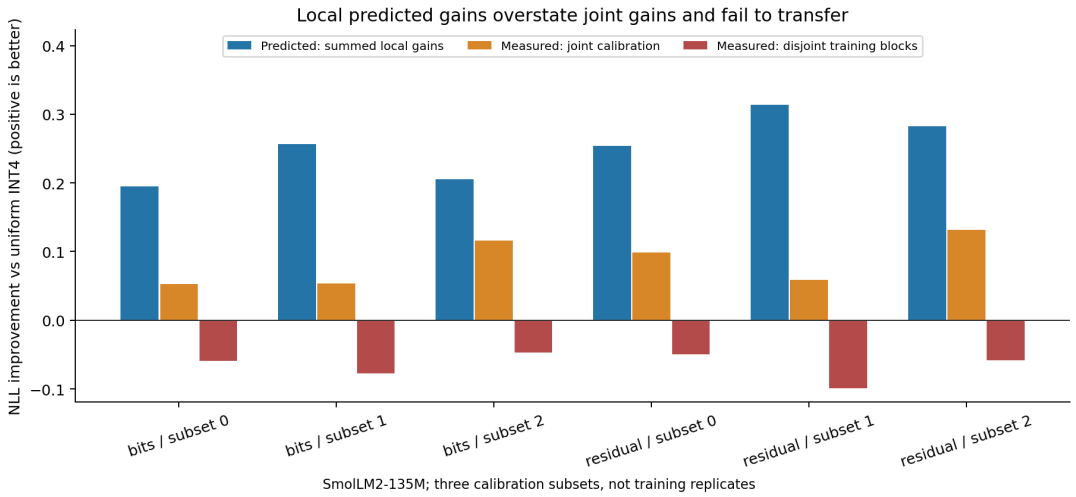


Figure 7. *Marginal allocation predictions overstate jointly realized gains.*

10. Storage, provenance and compute

Model	Target payload	Full checkpoint
SmolLM2-360M	176,947,200 B	271,561,031 B
Pythia-410M	198,844,416 B	302,615,137 B

Each model's seven Phase-4 arms match these byte counts exactly. The collection process verified all 52 expected artifacts and 42 dynamically selected checkpoint files, including checksums, source revision, model/data fingerprints, token counts, and the requirement that selection precedes holdout access.

Two private Kaggle runs completed on their first submission. Their runner intervals sum to approximately 1.511184 visible T4 device-hours, including installation but excluding export/platform overhead. This is execution telemetry, not exact account billing.

11. Reproducibility

Layer	Recorded evidence
Protocol	Frozen ADRs, append-only amendments and acceptance gates
Execution	Typed run specs and generated Kaggle/Colab notebooks
Environment	Python, PyTorch, Transformers, CUDA and hardware metadata
Storage	Payload/layout/full-checkpoint sizes plus checksums
Results	Original metrics, source scores and derived attestations
Publication	Artifact-generated tables/figures with recorded hashes
Quality gates	Formatting, types, CPU tests, docs and security CI

The Phase-4 scientific source was frozen at commit `71a1ec11cea6be90a6ab49356e81d980f4667784`. A new scientific claim requires a new run ID, new protocol and untouched data; collection scripts do not silently resubmit compute.

12. Limitations

- Calibration subsets are not independent model-training seeds.
- Only two sub-billion-parameter model families and two reserved corpora are tested.
- Perplexity uses dense reconstruction, not a deployed low-bit GPU kernel.
- The diagonal activation proxy omits feature covariance.
- Full matched GPTQ/AWQ arms were not run in Phase 4.
- Exposed holdouts cannot support further tuning; the next study needs fresh data.

13. Conclusions

SpectraQuant did not validate its original joint low-rank/low-bit allocator. Deployment-matched evaluation and joint-loss analysis made that negative result stronger. A simpler clipping pivot produced useful equal-byte gains, while the frozen transfer study showed that those gains depend on architecture, corpus and grouping.

What the study establishes

Activation-weighted INT4 clipping is a strong C4 control in the tested setting, with 5.75-7.70% lower perplexity across two architectures at identical storage. Its Pythia PG19 regression rules out a universal or domain-robust claim.

The broader research-engineering lesson is procedural: evaluate the deployed representation, freeze the next test, preserve negative cells, authenticate artifacts, and bound every claim by the evidence that actually ran.

References

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Artifact availability. Code, protocols, generated notebooks, bounded evidence, figures and LaTeX source are available at github.com/OussemaHarrabi/SpectraQuant.